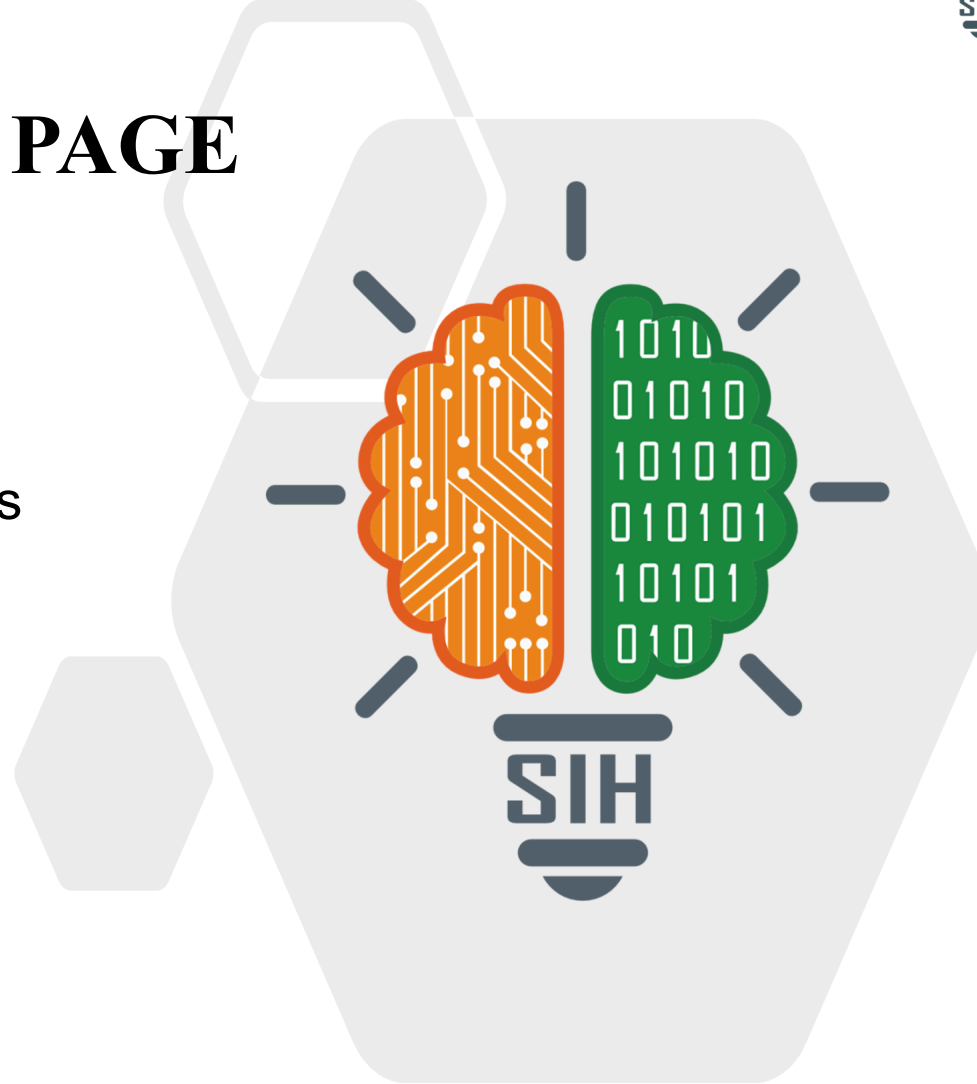


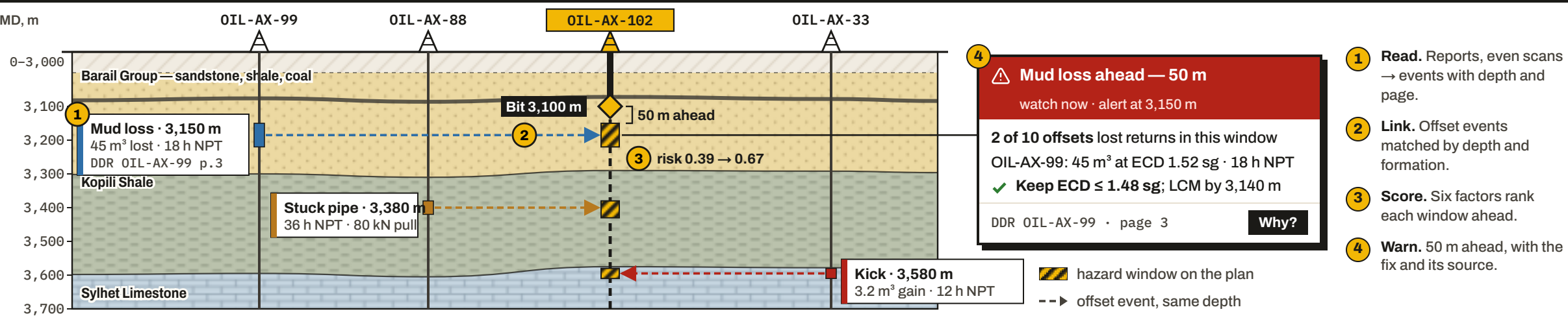
TITLE PAGE



Proposed solution. NWIS sits beside eRTMAC, links the bit depth to what nearby wells hit at that depth, and **warns before the bit gets there.**

Detailed explanation of the proposed solution

Demo field on real Upper Assam stratigraphy — illustrative wells, not Oil India records



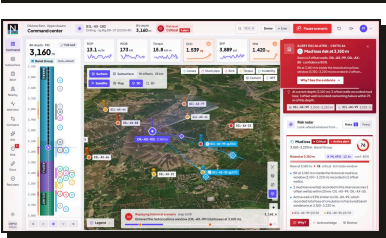
How it addresses the problem ✓ 20 / 20 official lines pass

The problem today	How NWIS addresses it
History sits in WCRs, DDRs, PDFs and people’s memory	one depth-indexed base; every event cited
Offset review is manual and slow	offset history at the bit depth, on a 3D map
Parameters, tops, casing and mud are seen apart	correlated across wells on one depth axis
The same losses, stuck pipe and kicks repeat	alert 50 m ahead, with the fix that worked

Innovation and uniqueness of the solution

- 01 **Depth is the key.** One bit depth drives the map, 3D view, risk, search and brief.
- 02 **Evidence first.** No answer without the report, page and depth behind it.
- 03 **Explainable, then learned.** A six-factor score, cross-checked by ML (AUC 0.86 vs 0.76).
- 04 **Built for rig routines.** Look-ahead brief for the 12-hour handover and DWOP; spoken alerts.
- 05 **Proven on real records.** 1,970 public well histories → 611 problems, 94 % precision.

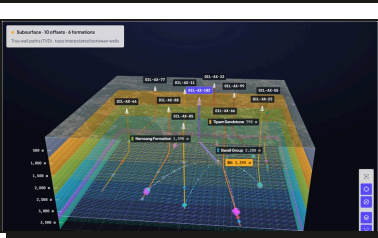
Working prototype



Alert with its evidence

 **Prototype**
<add link>

 **Video**
<add link>



Subsurface 3D at the bit

 **Source code**
github/aryabailur/sih26121

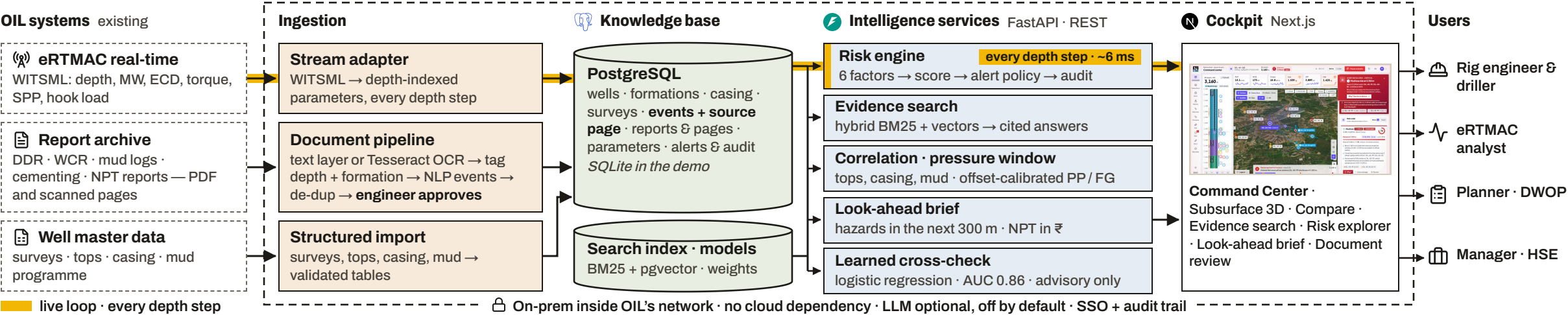
Technologies to be used

all open source · Claude only as an optional, grounded summariser

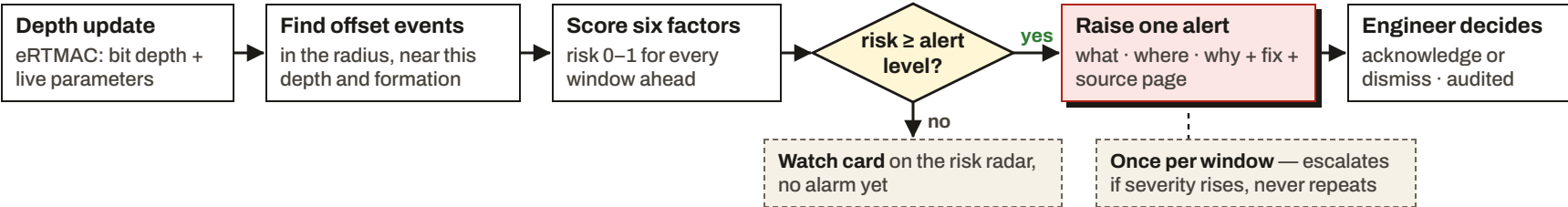
Frontend			Backend			AI · NLP · OCR			Data & integration		Quality	
Next.js 16	React 19	TypeScript	Tailwind CSS	Python 3	FastAPI	Tesseract OCR	Drilling NLP	BM25 + vectors	PostgreSQL	pgvector	pytest	
MapLibre GL	three.js	Recharts	Pydantic	SQLAlchemy	Logistic regr.	Claude (opt.)			SQLite	WITSML	Playwright	

Methodology and process for implementation

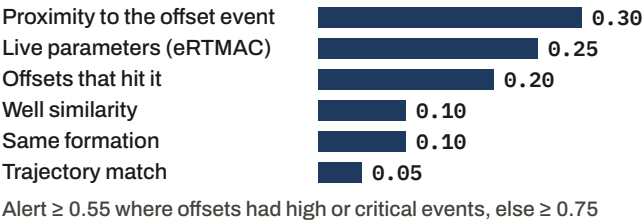
system architecture — a read-only layer beside eRTMAC



Every depth step — how an alert is raised



Risk score = weighted sum of six factors



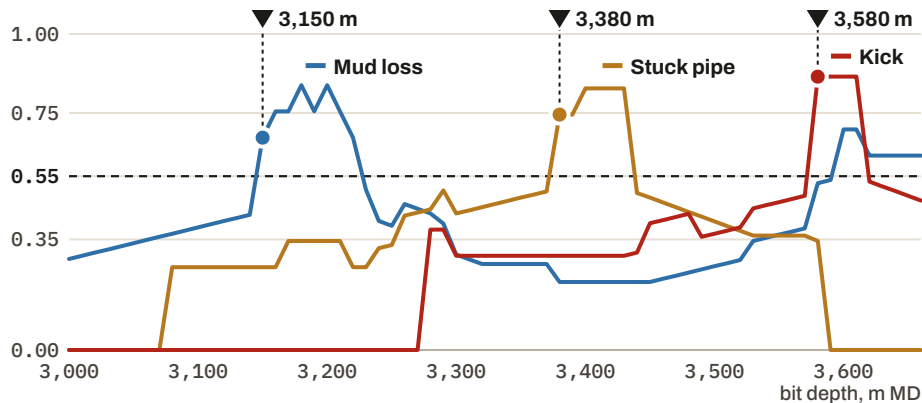
Analysis of the feasibility of the idea

built, tested, measured

Check	Result	Evidence
SIH26121 requirement lines	20 / 20 pass	API check
Automated tests	22 / 22 pass	pytest
Risk evaluation per depth step	~6 ms	median, in-process
Real well histories read	1,970 in 2.3 s	Sodir FactPages
Extraction precision, held-out	94 % (47 / 50)	hand-checked
Learned risk cross-check	AUC 0.86 vs 0.76	leave-one-well-out
Runs on	one on-prem server	no GPU, no licence

Replay of the demo well: alerts fire where the offsets struggled

OIL-AX-102, 3,000 → 3,660 m · ▼ alert raised · dashed line = alert level



Potential challenges and risks → strategies for overcoming these challenges

Challenge / risk	Level	Strategy	Status
Confidential well data	High	On-prem inside OIL's network, no cloud dependency; LLM optional and off by default; role-based access and an audit trail.	Built
Old, scanned, inconsistent reports	High	Tesseract OCR with per-page confidence, a drilling synonym lexicon, and an engineer approves every extracted event.	Built
Alert fatigue on the rig	Medium	Weak windows need live confirmation; one alert per window that escalates, never repeats.	Built
Trust in AI advice	Medium	Every alert opens Why? — factors, cited evidence and confidence. The engineer decides.	Built
Weights tuned on a demo field	Medium	Refit on OIL's NPT history, then a shadow pilot before any alert goes live.	Pilot
Plugging into eRTMAC	Low	Read-only WITSML adapter into the same parameter schema the prototype already uses.	Pilot

Viability — from prototype to OIL's rigs in 12 months

no licence fee · one saved NPT day ≈ ₹30 lakh

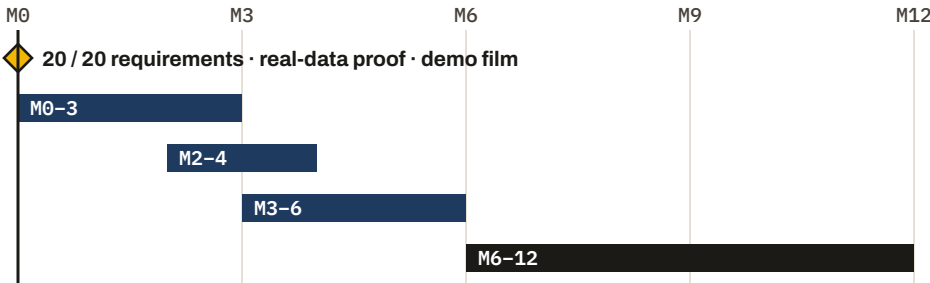
- Prototype built and verified

Ingest one OIL field (DDR / WCR archive)

Refit the weights on OIL's NPT history

Shadow pilot at eRTMAC, 2 rigs, silent

Go live: WITSML, SSO, all Assam rigs



Potential impact on the target audience

Who	Today	With NWIS
 Rig engineer & driller	reads offset reports by hand before each section	sees what offsets hit at this depth — 50 m ahead, with the fix that worked
 eRTMAC monitoring team	live curves without the field's history	curves judged against offset limits, e.g. the loss-onset ECD
 Planners & DWOP	days of offset study per well plan	a look-ahead brief in seconds: hazards, mud window, NPT in ₹
 New engineers	expertise retires with the people	a searchable field memory where every line is cited
 Managers & HSE	NPT is counted after it happens	expected NPT per section before drilling; every alert audited

Benefits of the solution

₹ Economic

Fewer repeated losses, stuck pipe and kicks → less NPT and rig time; faster well planning.

🌿 Environmental

Fewer loss events → less mud, LCM and cement lost downhole; fewer rig-days of diesel.

🛡️ Safety

Earlier awareness of kicks and overpressure supports well control and crew safety.

👤 Social & strategic

Keeps OIL's drilling know-how in-house; supports India's push for more domestic oil and gas.

Why it matters — where drilling time goes

Well-construction time



Inside NPT: kicks, lost circulation and wellbore instability



↑ exactly the events NWIS warns about, well by well and depth by depth

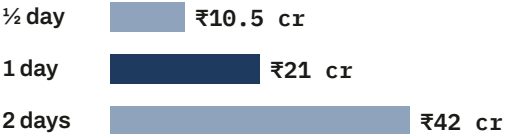
Value at OIL's scale

≈ ₹21 crore a year

70 wells a year [4] × 1 NPT day saved per well × ₹30 lakh a rig-day*

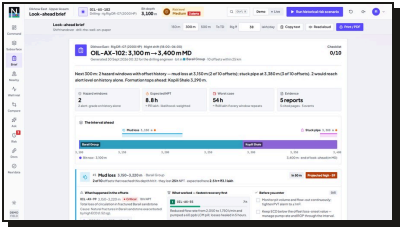
* assumed spread rate — an illustration, not a forecast; editable in the brief.

If NWIS saves per well...



Priced per section, before drilling

prototype · demo well



The look-ahead brief for OIL-AX-102 at 3,100 m: 2 hazard windows in the next 300 m, 8.8 h expected NPT ≈ ₹11 lakh, worst case 54 h ≈ ₹68 lakh — with what worked in the offsets, ranked by the NPT it saved. Printable for the 12-hour handover.

Research — where today’s approaches fall short

Capability	Manual offset review	Real-time monitoring alone	Generic document chatbot	NWIS
Tied to the live bit depth	☐	☒	☐	☒
Reads DDR / WCR text, even scans	☒	☐	☐	☒
Cites report, page and depth	☐	☐	☐	☒
Warns before the risky depth	☐	☐	☐	☒
Time to the offset history	hours – days	—	seconds	< 1 s, cited

☒ yes ☐ partly ☐ no

Earlier work mines report text [5] or predicts stuck pipe from drilling parameters [6] — separately. NWIS joins both at the live bit depth and shows the source of every warning.

Details / links of the reference and research work

Problem & industry data

- [1] SIH26121 Nearby Wells Intelligence System — Oil India Limited — [SIH 2026 portal](#)
- [2] NPT is about 20 % of well-construction time — [Drilling Contractor, 2018](#)
- [3] Kicks, losses and wellbore instability: up to 40 % of NPT — [Offshore Magazine, 2011](#)
- [4] OIL spudded 70 wells in FY25 with 21 rigs — [Chairman’s speech, 66th AGM, 2025](#)

Research, data & standards

- [5] Hoffmann et al. — pattern mining in drilling reports with deep NLP — [arXiv, 2017](#)
- [6] Montes, Ashok, van Oort — review of stuck-pipe prediction — [SPE Journal, 2025](#)
- [7] Norwegian Offshore Directorate well data, NLOD 2.0 — [Sodir FactPages](#)
- [8] WITSML real-time drilling data standard (eRTMAC adapter) — [Energistics](#)

Validated on real public well records

same extractor, rules frozen

1,970

well histories read, 1966–2022

611

drilling problems found in 2.3 s

94 %

precision on held-out wells (47 of 50)

Problems found, by type — Norwegian shelf [7]

